

Longitudinal Cardiac Progression Prediction From ICU Records: Comparing Sequence Models and Temporal Feature Gradient Boosting

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Abstract—Cardiovascular deterioration is a longitudinal process, yet many predictive models summarize a patient with a single encounter and therefore underuse trajectory information available in intensive-care records. This paper studies binary prediction of worsening cardiac progression from longitudinally ordered ICU stays in MIMIC-IV version 2.1. We construct a cardiac cohort from ICD diagnosis codes, aggregate vital and laboratory measurements into 59 features per stay, form fixed-length six-visit sequences, and define worsening using a Cardiac Severity Index (CSI) increase of more than 10% between consecutive stays, with in-hospital mortality as a fallback label for terminal observations. Two modeling tracks are compared: (i) end-to-end sequence classifiers (bidirectional LSTM with attention, bidirectional GRU, and a Transformer encoder) and (ii) gradient-boosting models trained on explicit temporal descriptors of change, variability, and acceleration, followed by isotonic calibration and validation-based threshold selection. On a held-out test set of 5,289 sequences (16.9% positive), a reproduced Transformer baseline achieved $F1 = 0.6319$ and $ROC-AUC = 0.9168$, whereas a calibrated soft-voting ensemble of HistGradientBoosting, LightGBM, and XGBoost achieved $F1 = 0.7334$, precision = 0.7226, recall = 0.7444, and $ROC-AUC = 0.9462$. For short ICU trajectories, explicit temporal feature engineering with calibrated boosting improved the precision-recall operating point relative to larger sequence models in this internal evaluation. The study is retrospective research and is not intended for diagnosis or clinical decision-making.

Index Terms—MIMIC-IV, cardiac progression, longitudinal electronic health records, temporal feature engineering, gradient boosting, Transformer, ICU risk prediction.

I. INTRODUCTION

Cardiovascular diseases remain a leading cause of global mortality and disability [1]. In critical care, deterioration is rarely instantaneous: vital signs, cardiac biomarkers, and metabolic laboratories evolve across successive intensive-care unit (ICU) encounters. Models that encode only a single snapshot can miss the direction, speed, and instability of that evolution. Longitudinal electronic health records (EHRs) therefore motivate progression-oriented prediction rather than static disease classification [2]–[4].

Prior clinical machine-learning work has achieved strong results for mortality, decompensation, phenotype labeling, and next-visit code prediction using recurrent and attention-based architectures [2], [3], [5], [6]. Parallel work shows that gradient-boosted trees remain competitive for structured clinical features [7], [8]. What remains less clear for short

ICU trajectories is whether end-to-end sequence models automatically learn the clinically relevant temporal structure, or whether making change, variability, and acceleration explicit yields a better precision-recall tradeoff under class imbalance.

This paper addresses that question on a cardiac cohort derived from MIMIC-IV 2.1 [9], [10]. We formulate worsening progression prediction as a patient-trajectory classification task over chronologically ordered ICU stays, compare bidirectional recurrent and Transformer sequence models with a temporal-feature gradient-boosting track, and evaluate discrimination and operating-point metrics under an approximately 16.9% positive-class prevalence. The central finding is empirical rather than architectural: on six-visit sequences, calibrated boosting over engineered temporal descriptors outperformed the strongest reproduced sequence baseline in F1 and ROC-AUC while remaining compact for CPU inference.

The main contributions are:

- 1) a reproducible formulation of binary cardiac progression prediction on MIMIC-IV 2.1 using six-visit ICU sequences with 59 features per visit and a CSI-based worsening label;
- 2) a controlled comparison of BiLSTM-attention, BiGRU, and Transformer sequence classifiers against HistGradientBoosting, LightGBM, XGBoost, and calibrated ensembles trained on explicit temporal descriptors;
- 3) quantitative evidence that, for these short sequences, temporal feature engineering with isotonic calibration and validation-tuned thresholds improved F1 from 0.6319 to 0.7334 and ROC-AUC from 0.9168 to 0.9462 relative to a reproduced Transformer baseline; and
- 4) an explicit audit of methodological risks—including sample-level rather than patient-disjoint splitting, label heterogeneity from mortality fallback, and proxy medication features—that constrain clinical interpretation.

II. RELATED WORK

A. Cardiovascular Risk and Progression Modeling

Classical cardiovascular risk tools emphasize long-horizon population risk using sparse baseline covariates. More recent EHR studies model recurrent heart-failure trajectories and comorbidity-conditioned progression with recurrent networks [11]. ECG-centric deep models such as SCN-Deep BiL-

STM capture spatial–temporal waveform patterns for heart-disease prediction [12], while deep classifiers with SHAP explanations have been applied to heart-attack risk on comparatively static tabular inputs [13]. Those settings differ from short, high-resolution ICU stay sequences with a physiology-derived severity change label.

B. Machine Learning on Electronic Health Records

Public critical-care databases, especially MIMIC and the broader PhysioNet ecosystem, have enabled reproducible clinical machine learning [3], [9], [14], [15]. Benchmark tasks commonly include in-hospital mortality, physiologic decompensation, length of stay, and phenotype classification [3]. Large-scale EHR deep learning further shows that sequential clinical events can support scalable prediction [4]. Such efforts demonstrate the value of longitudinal physiologic time series, but they do not directly define cardiac progression as a CSI transition between successive ICU stays. Our cohort construction and label definition are therefore complementary to standard MIMIC benchmarks.

C. Temporal and Sequential Clinical Modeling

Recurrent models such as Doctor AI predict subsequent diagnoses and medications from visit histories [2]. Attention mechanisms, including RETAIN, improve interpretability by highlighting influential visits and codes [5]. LSTM architectures for clinical time series established early evidence that sequential deep models can learn diagnostic structure from irregular physiologic streams [6]. Transformer self-attention further removed recurrence as a requirement for sequence transduction [16]. Recent longitudinal work applies Transformers to heart-failure detection from in-patient EHR trajectories [17], evolving health trajectories [18], outcome-guided deep temporal clustering of progression subtypes [19], and explainable early chronic-disease prediction from longitudinal EHRs [20]. These methods are well suited to longer code or multi-year sequences; whether their capacity is necessary for six-step ICU trajectories with dense numeric features remains an empirical question addressed here.

D. Gradient Boosting for Structured Clinical Prediction

XGBoost and LightGBM are widely used for nonlinear tabular prediction because they capture feature interactions efficiently and train quickly relative to deep models [7], [8]. In clinical settings they are often paired with hand-crafted aggregates and implemented with mature scientific Python tooling [21]. Our temporal-feature track follows that tradition: we do not propose a new boosting algorithm, but rather evaluate whether descriptors of trajectory shape make short-sequence progression prediction more amenable to calibrated tree ensembles [22].

E. Literature Comparison

Table I summarizes representative comparison works and their limitations relative to the present ICU progression task. Incomplete or unverifiable titles that appear in some slide

decks (for example, incompletely titled Zangana heart-disease studies) are omitted; we cite only the verified explainable-Transformer EHR paper [20] together with the other Crossref-checked references above.

F. Research Gap

Prior work establishes strong ECG and static cardiac classifiers [12], [13], longitudinal Transformer EHR models [17], [18], [20], and temporal clustering of progression patterns [19], alongside mature MIMIC resources [9]. Fewer studies directly ask, under a fixed short ICU trajectory length ($L = 6$), a CSI-based cardiac progression label, and approximately 17% class imbalance, whether explicit temporal descriptors plus calibrated boosting can outperform end-to-end sequence learners. That dual-track comparison—and the accompanying leakage, label, calibration, and thresholding analysis—is the gap addressed by this paper.

III. PROBLEM FORMULATION

Let patient i contribute a chronologically ordered sequence of ICU-stay feature vectors

$$X_i = \{x_{i,1}, x_{i,2}, \dots, x_{i,T_i}\}, \quad x_{i,t} \in \mathbb{R}^d, \quad (1)$$

where $d = 59$ in this study. A fixed-length window of $L = 6$ consecutive stays is extracted (with left padding when $T_i < 6$). For a window ending at stay t , the binary target is

$$y_{i,t} = \begin{cases} 1, & \text{if } CSI_{i,t+1} > 1.10 CSI_{i,t}, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

when a subsequent stay exists. For a patient’s final or only stay, the implementation substitutes the hospital in-hospital mortality flag. Thus $y = 1$ denotes predicted worsening progression under a heterogeneous operational definition: CSI increase for transitions and mortality for terminal observations. The learning task is to estimate $p(y = 1 \mid X)$ and choose a decision threshold on validation data.

IV. MATERIALS AND METHODS

A. Dataset and Cohort Construction

Experiments use the deidentified MIMIC-IV version 2.1 hospital and ICU modules [9], [10], [14]: `patients`, `admissions`, `diagnoses_icd`, `icustays`, `chartevents`, and `labevents`. Access to MIMIC-IV requires PhysioNet credentialing and a data-use agreement; the present analysis uses only deidentified records.

A cardiac cohort is selected using ICD-9 prefixes 410–414, 428, 426–427, 425, 394–397, 424, 402, 404, 415–417, and 429, together with ICD-10 prefixes I21–I25, I50, I44–I49, I42, I34–I37, I11, I13, I26, I27, and I51. These prefixes cover ischemic heart disease, myocardial infarction, heart failure, arrhythmia, cardiomyopathy, valvular disease, hypertensive heart disease, pulmonary heart disease, and related conditions. Patient age uses MIMIC `patients.anchor_age`; no additional disease-specific age exclusion is applied beyond the ages available in the released tables.

TABLE I
COMPARISON OF RELATED WORK VERSUS THE PRESENT SHORT-SEQUENCE ICU CARDIAC PROGRESSION TASK.

Ref.	Year	Method	Data focus	Strength	Limitation relative to our task
[12]	2025	SCN-Deep BiLSTM	ECG waveforms	Spatial-temporal feature learning	ECG signals; not ICU stay progression
[13]	2024	DL + SHAP	Heart-attack tabular risk	Strong accuracy with local explanations	Largely static inputs; limited trajectory modeling
[17]	2024	Transformer	Longitudinal in-patient EHR / HF	Captures temporal HF dynamics	Different label/task; heavier data needs
[9]	2023	Dataset release	MIMIC-IV hospital/ICU EHR	Enables reproducible critical-care ML	Resource paper, not a progression predictor
[18]	2025	Evolve Transformer	Longitudinal health trajectories	Models evolving multi-visit trajectories	Not CSI-based short ICU cardiac progression
[19]	2024	OG-DTC clustering	Longitudinal disease progression	Outcome-guided temporal subtypes	Clustering focus; limited bedside deployment form
[20]	2026	Explainable Transformer	Longitudinal chronic-disease EHR	Temporal prediction with explainability	Chronic-disease setting; not 6-visit ICU CSI labeling

The reported full run identified 66,841 cardiac subjects, 157,019 cardiac admissions, and 52,130 cardiac ICU stays before sequence construction. After preprocessing and six-visit window generation, the model-ready dataset contained 35,256 samples: 24,678 training, 5,289 validation, and 5,289 testing samples. Each split contained approximately 16.9% positives. The held-out test set contained 896 worsening and 4,393 stable/improving examples.

B. Feature Extraction and Preprocessing

Vital signs are streamed from `chartevents` and laboratory values from `labevents`. Extracted variables include heart rate; systolic, diastolic, and mean blood pressure; respiratory rate; temperature; oxygen saturation; fingerstick glucose; troponin; BNP/NT-proBNP; creatinine; BUN; glucose; electrolytes; bicarbonate; hematocrit; hemoglobin; WBC; and platelets. Repeated measurements within a stay are summarized using statistics such as mean, standard deviation, and maximum. Contextual variables include age, length of stay, prior myocardial infarction, prior heart failure, prior arrhythmia, diagnosis counts, admission type, time since the previous admission, time from the first ICU stay, and visit index.

The resulting representation contains 59 features per ICU stay. Missing values are forward- and backward-filled within each patient timeline; remaining missing values are imputed with training-split feature medians, with zeros used for any residual non-finite values. A `StandardScaler` is fit on the training split only and applied to validation and test data. The final tensor has shape $(N, 6, 59)$.

C. Cardiac Severity Index and Label Semantics

CSI is a hand-engineered severity score assembled from abnormalities in heart rate, systolic blood pressure, Troponin T, BNP or NT-proBNP, SpO_2 , creatinine, respiratory rate, and hemoglobin. Each component contributes a bounded non-negative increment (typically capped near 1.5–3 points depending on the variable), so higher CSI indicates greater instantaneous severity. Because CSI is not a clinically adjudicated gold standard, all progression claims in this paper refer to the operational CSI/mortality label rather than to chart-reviewed deterioration.

D. Sequence Construction, Splitting, and Leakage Controls

Patients are ordered chronologically. Sliding windows of six visits are created for patients with sufficient history; shorter histories are left-padded by repeating the earliest available row. Samples are partitioned 70%/15%/15% with stratified random splitting and `random_state=42`. Scaling uses training data only.

a) *Patient-level contamination.*: The current implementation splits generated windows rather than enforcing patient-disjoint partitions. Overlapping windows from the same patient may therefore appear across splits, which can inflate apparent generalization. Results should be interpreted as an internal benchmark pending patient-disjoint and chronological evaluation.

b) *Prediction-time leakage.*: Labels for transitional samples use CSI_{t+1} relative to CSI_t ; features for a window are constructed from stays available at prediction time within that window. Future diagnoses after the prediction point, discharge disposition beyond the label definition, and later laboratory values outside the window must not enter the feature set. Medication-count features in the current pipeline are documented as proportional proxies derived from diagnosis counts rather than from prescription tables; those proxies should be replaced with direct medication extraction in future revisions.

E. Sequence Models

Three neural baselines consume the raw (6×59) sequence:

- **BiLSTM-attention**: two-layer bidirectional LSTM (hidden size 128, dropout 0.3) with attention pooling over visit representations and a fully connected classification head.
- **BiGRU**: two-layer bidirectional GRU with the same hidden size and dropout, using the final bidirectional state as the sequence representation.
- **Transformer encoder**: each visit is projected to $d_{\text{model}} = 128$; a learnable classification token is prepended; sinusoidal positional encoding is added; three encoder layers with eight heads, feed-forward width 512, and dropout 0.1 produce a sequence representation for a two-layer classification head [16].

Neural models use weighted binary cross-entropy with logits to address class imbalance, AdamW optimization, gradient clipping, validation-loss checkpointing, and early stopping. Reproduced fair-comparison runs used learning rate 3×10^{-4} , batch size 128, up to 80 epochs, and patience 12, implemented in PyTorch [23]. Reported neural operating thresholds in the baseline tables are 0.5 unless otherwise stated.

F. Temporal Feature Gradient Boosting

“Temporal gradient boosting” in this work denotes gradient boosting trained on temporally engineered tabular features; it is not a new boosting algorithm. Each (6×59) sequence is converted into a flat descriptor vector containing the flattened sequence together with per-feature summaries: first and last values, mean, standard deviation, minimum, maximum, median, interquartile range, range, delta, coefficient of variation, slope, half-split trend, consecutive-change velocity, second-difference acceleration, total variation, and position of the last value within the observed range. A “simple” configuration uses 826 features; richer configurations feed HistGradientBoosting, LightGBM, and XGBoost [7], [8], [21].

Candidates are selected using validation PR-AUC. Selected models are calibrated with isotonic regression on validation predictions [22]. A decision threshold maximizing validation F1 is then frozen and applied to the test set. The soft-voting ensemble averages calibrated probabilities from HistGradientBoosting, LightGBM, and XGBoost; its documented operating threshold is 0.335. A stacked ensemble is also evaluated. Selected backend hyperparameters from the reported improved run include HistGradientBoosting with learning rate 0.08, maximum leaf nodes 63, and minimum samples per leaf 20; LightGBM with learning rate 0.05, 63 leaves, and minimum child samples 20; and XGBoost with learning rate 0.03, maximum depth 6, and subsample 0.8.

G. Evaluation Metrics

Because positives comprise only $\approx 16.9\%$ of samples, accuracy alone is insufficient. We report accuracy, precision, recall (sensitivity), F1, ROC-AUC (tables: AUC), and PR-AUC / average precision (tables: AP). Precision and recall are defined with respect to the worsening class. For imbalanced progression prediction, PR curves and F1 at a validation-chosen threshold are particularly informative; ROC-AUC remains useful for ranking quality, but the precision–recall view is emphasized under imbalance [24], [25].

H. Experimental Setup

Primary neural training was performed in Google Colab on an NVIDIA Tesla T4 GPU; boosting models were trained on CPU. The software stack uses Python with scikit-learn, PyTorch, LightGBM, and XGBoost as specified in the project dependency files [21], [23], [26]. Stratified splits use `random_state=42`. Neural results in Table II come from a full-run training pass; Table III reports a later reproduction on the identical arrays used by the boosting track and is the fair baseline for improvement claims. Multi-seed means

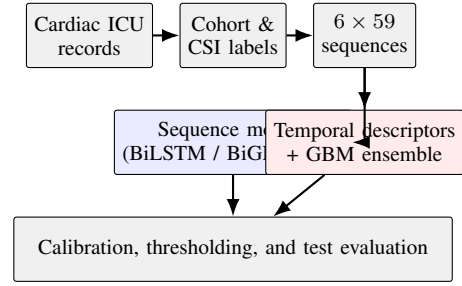


Fig. 1. Overview of the dual-track pipeline: sequence models consume raw six-visit tensors, while the boosting track consumes engineered temporal descriptors.

TABLE II
ORIGINAL FULL-RUN SEQUENCE-MODEL RESULTS ON THE HELD-OUT TEST SET (THRESHOLD 0.5).

Model	Acc.	Prec.	Rec.	F1	AUC
BiLSTM-Attn	0.8134	0.4712	0.8304	0.6012	0.9021
BiGRU	0.8132	0.4712	0.8404	0.6038	0.9076
Transformer	0.8198	0.4814	0.8237	0.6077	0.9096

and confidence intervals are not available for the reported tables and are therefore listed as a limitation rather than as unpublished results.

A research-prototype demonstration accompanying the codebase uses synthetic patient scenarios to avoid exposing protected health information; all numeric claims in this paper are taken from held-out MIMIC evaluation files rather than from the demo.

V. RESULTS

A. Neural Sequence Baselines

Table II summarizes the original full-run sequence results at threshold 0.5. The Transformer obtained the best accuracy, precision, F1, and ROC-AUC; BiGRU obtained the highest recall. Weighted loss produced high recall with precision near 0.48, illustrating an aggressive alert profile at the default threshold.

Table III reports the fair reproduction on the arrays shared with the boosting experiments. The reproduced Transformer is the primary sequence baseline for subsequent comparisons.

B. Temporal Feature Gradient-Boosting Results

Table IV reports calibrated boosting results at validation-selected thresholds. The soft-voting ensemble achieved the best F1 (0.7334) and tied the best ROC-AUC (0.9462). The stacked ensemble slightly improved accuracy, precision, and PR-AUC at the cost of lower recall and F1. Individual HistGradientBoosting, LightGBM, and XGBoost models remained close to one another, indicating that the temporal representation—not a single backend—drove most of the gain.

Relative to the reproduced Transformer, the soft-voting ensemble improved F1 by 0.1015 absolute and ROC-AUC by 0.0294, while raising precision from 0.5174 to 0.7226 and retaining recall 0.7444. Fig. 2 summarizes that comparison.

TABLE III
REPRODUCED SEQUENCE BASELINES ON THE ARRAYS USED BY THE BOOSTING TRACK (THRESHOLD 0.5).

Model	Acc.	Prec.	Rec.	F1	AUC
BiLSTM-Attn	0.8308	0.5003	0.8103	0.6187	0.9060
BiGRU	0.8211	0.4837	0.8259	0.6101	0.9099
Transformer	0.8399	0.5174	0.8114	0.6319	0.9168

TABLE IV
TEMPORAL-FEATURE GRADIENT-BOOSTING RESULTS ON THE HELD-OUT TEST SET (VALIDATION-CALIBRATED THRESHOLDS).

Method	Acc.	Prec.	Rec.	F1	AUC	AP
Soft vote	0.9083	0.7226	0.7444	0.7334	0.9462	0.8302
Stacked	0.9125	0.7612	0.7042	0.7316	0.9462	0.8307
HistGB	0.9126	0.7747	0.6830	0.7260	0.9434	0.8093
LightGBM	0.9079	0.7332	0.7176	0.7253	0.9439	0.8153
XGBoost	0.9032	0.6983	0.7545	0.7253	0.9433	0.8069
Simple (826)	0.9102	0.7451	0.7143	0.7293	0.9450	0.8115

The gain is therefore not explained by simply predicting more positives.

Fig. 3 shows ROC and precision–recall curves for sequence and boosting models. Boosting curves dominate across most of the operating range. Primary quantitative claims in this paper rely on the tables; figures are retained for qualitative comparison and should be read as illustrative when legend annotations differ from tabulated metrics.

C. Error Profile and Feature Attribution

At threshold 0.335, the soft-voting ensemble confusion matrix on the test set counts 4,137 true negatives, 667 true positives, 256 false positives, and 229 false negatives (Fig. 4, left). Feature-gain attributions from LightGBM highlight oxygen-saturation variability, bicarbonate, blood-pressure and heart-rate levels or variability, age, length of stay, temperature, and WBC-related descriptors among the leading signals (Fig. 4, right). These associations describe model behavior; they are not causal effects and should not be interpreted as treatment guidance.

VI. DISCUSSION

For six-visit ICU trajectories, the dominant signal appears to be short-horizon change rather than long-range sequential dependence. Explicit deltas, slopes, velocities, accelerations, and variability statistics expose that structure to tree ensembles, which then model nonlinear interactions efficiently. Sequence models can in principle learn analogous representations, but under the present architecture, loss weighting, and fixed threshold of 0.5, they favored recall over precision.

Calibration and threshold selection are first-class design choices [22]. Fig. 5 illustrates how operating points move along the precision–recall frontier as the threshold changes. In any prospective decision-support setting, thresholds should be chosen against an explicit utility or workload constraint rather than F1 alone.

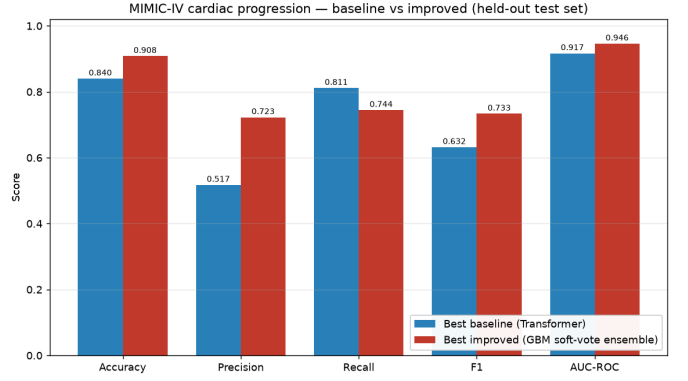


Fig. 2. Held-out comparison of the reproduced Transformer baseline and the GBM soft-voting ensemble.

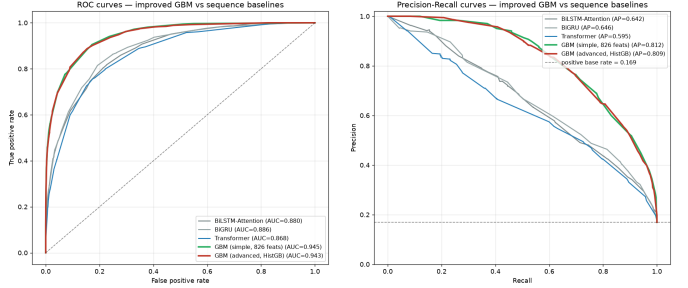


Fig. 3. ROC and precision–recall curves for sequence baselines and temporal-feature boosting models. Prefer Table metrics when legend values conflict.

Deployment considerations also favor the boosting track in this study: the documented deployable artifact is approximately 3.3 MB and supports CPU inference without a deep-learning runtime [26]. That advantage is practical rather than clinical; compact artifacts do not imply bedside readiness.

VII. LIMITATIONS

Several limitations constrain interpretation. First, evaluation is internal to one center’s deidentified database and uses sample-level rather than patient-disjoint splits. Second, CSI is hand-engineered and not externally validated against clinician-adjudicated progression. Third, mortality fallback creates label heterogeneity between transitional and terminal samples. Fourth, missingness indicators are not yet explicit inputs, and medication-count features are proxy-derived. Fifth, subgroup performance by age, sex, race, admission type, and cardiac subtype is unreported. Sixth, statistical significance testing, calibration-error summaries, and multi-seed variability with confidence intervals are not available for the reported results. Seventh, improved deep-learning recipes and a proposed gated temporal attention network are documented in the project repository but lack finalized numbers suitable for inclusion here [26]. Future work should also include ablations that isolate static versus temporal descriptors and CSI-only versus mortality-fallback labels, as well as classical baselines such as logistic regression and random forests on the same temporal features under patient-disjoint chronological evaluation.

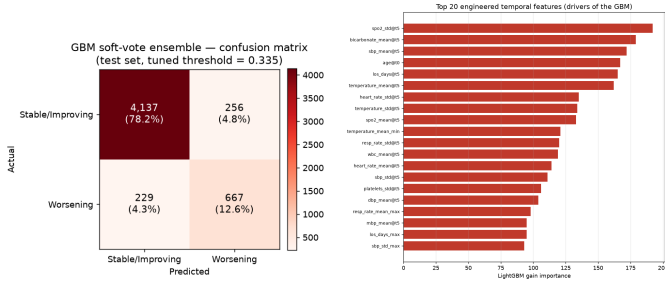


Fig. 4. Confusion matrix for the GBM soft-voting ensemble and leading LightGBM gain features.

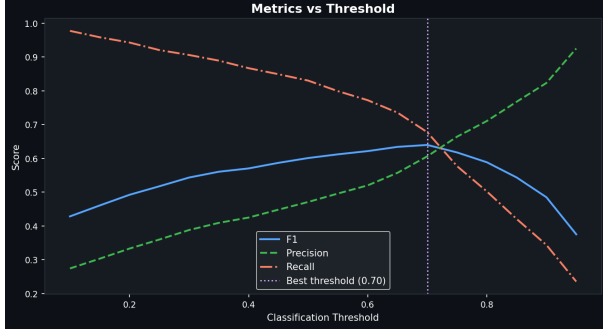


Fig. 5. Example threshold sweep for an LSTM/BiLSTM model, illustrating the precision–recall tradeoff as the decision threshold varies.

VIII. ETHICAL AND DATA-USE CONSIDERATIONS

MIMIC-IV is a deidentified critical-care database released through PhysioNet [9], [10], [14]. Use requires credentialing and acceptance of the applicable data-use agreement. This paper reports a retrospective methodological study only. No IRB approval number is claimed here beyond the conditions attached to MIMIC-IV access, and the models are not approved for clinical use. Any interactive demonstration of the research prototype uses synthetic scenarios for PHI safety and must not be confused with the held-out MIMIC metrics reported above.

IX. REPRODUCIBILITY

Cohort construction, sequence building, model training, and visualization scripts are available in the project repository [26]. Raw MIMIC-IV files cannot be redistributed; credentialed users must obtain them from PhysioNet. Preprocessed arrays, the stratified-split seed used for the reported partition, evaluation scripts, and the Python dependency specifications in `requirements.txt` should be sufficient for re-training once local MIMIC access is configured.

X. CONCLUSION

This paper formulated longitudinal cardiac progression prediction on MIMIC-IV 2.1 six-visit ICU sequences and compared end-to-end sequence models with calibrated gradient boosting over engineered temporal descriptors. The reproduced Transformer baseline reached F1 0.6319 and ROC-AUC 0.9168; a soft-voting HistGradientBoosting–LightGBM–XGBoost ensemble reached F1 0.7334 and ROC-AUC 0.9462

on the same held-out test set. The results support the hypothesis that explicit short-horizon temporal structure can be more effective than larger sequence architectures for this task setting, subject to the stated leakage, label, and single-dataset limitations. Future work should enforce patient-disjoint chronological evaluation, validate labels clinically, report subgroup and calibration analyses, and complete pending deep-model experiments before any translational claim is considered.

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